

GE261004 DESIGN THINKING

Unit IV – Prototyping and Testing · Presentation 1 of 6

Concept Visualisation and Why We Prototype

Build to think · the cost of changing your mind late · what a prototype is for · role, look and feel, implementation · choosing a fidelity

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In this presentation

- Build to think
- Why early and rough
- What a prototype is for
- What are you prototyping?
- Three questions a prototype answers
- Choosing a fidelity
- One prototype, one question
- Good and bad prototypes

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Why prototype at all

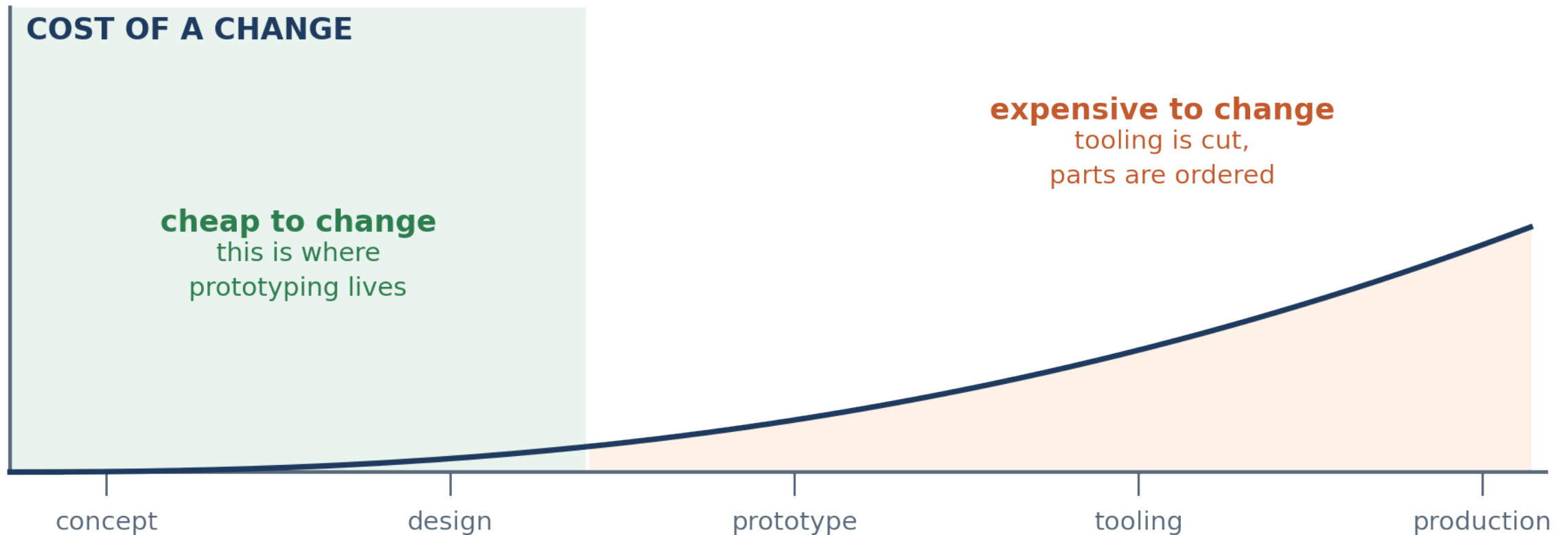
Thinking with your hands, while changing your mind is still cheap

Build to think

- A prototype is not the last step of designing – it is part of the thinking
- Making something forces decisions that discussion lets you avoid
- It moves the argument from opinion to evidence: the thing either works or it does not
- And it gives a user something to react to, which they cannot do to a description

KEY POINT You do not build a prototype because you have decided. You build one because you have not.

Why early and rough



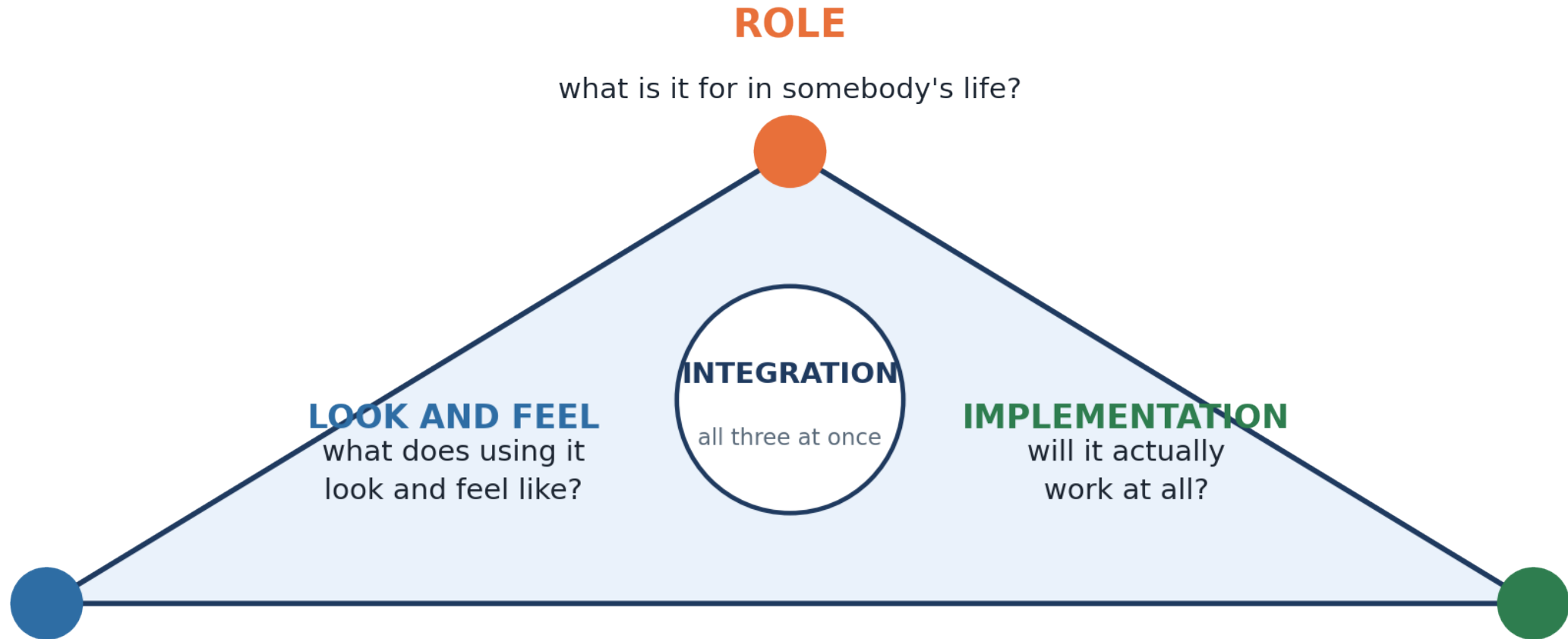
Every week you delay a change, it costs more to make.

What a prototype is for

- To answer one question you cannot answer by arguing
- To find the flaw while it is still free to fix
- To communicate an idea to people who do not read drawings
- To get permission – something people can hold is easier to fund than a slide

KEY POINT A prototype is an experiment with a question attached. No question, no prototype.

What are you prototyping?



Houde and Hill, Apple, 1997 – name the dimension before you build.

Three questions a prototype can answer

Role

Does this belong in the user's life at all?
Tested with storyboards, role play, a fake service.

Look and feel

What is it like to use?
Tested with mock-ups, models you can hold, click-throughs.

Implementation

Can it be made to work? Tested with rigs, circuits, code – ugly is fine here.

KEY POINT Most failed prototypes tried to answer all three at once, and answered none of them well.

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Choosing a fidelity

How finished does it have to be to teach you something?

The fidelity spectrum

LOW FIDELITY

HIGH FIDELITY

Sketch

Paper mock-up

Working model

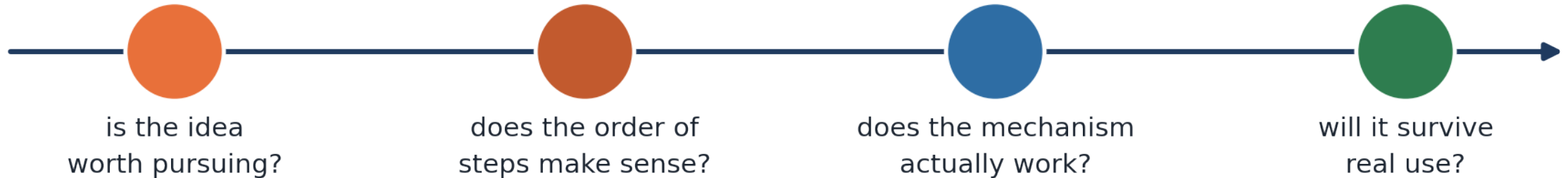
Production-like

minutes

an hour

days

weeks



go only as high as the question needs — fidelity you did not need is time you did not have

Each level costs more and answers a narrower question.

Fidelity and what it buys you

Level	Time	Answers	Risk
Sketch	Minutes	Is this worth pursuing?	Too vague to test
Paper mock-up	An hour	Does the sequence work?	Feels unreal
Working model	Days	Does the mechanism work?	You get attached
Production-like	Weeks	Will it survive use?	Too late to change

One prototype, one question

- Write the question on the prototype itself, in marker, before you start
- Build only what that question needs – everything else is decoration
- Two questions means two prototypes, not one bigger one
- If you cannot say what would make it fail, you do not have a question yet

KEY POINT “We built a prototype” is not a result. “We learned that users never noticed the second button” is.

Good and bad prototypes

A good prototype

- Answers one stated question
- Is as rough as it can be
- Is built in hours, not weeks
- You are willing to throw away

A bad prototype

- Shows off everything at once
- Is polished before it is tested
- Took so long you must defend it
- Nobody outside the team has held

KEY POINT The more time a prototype took, the less willing you will be to hear that it is wrong.

In class: name your question

- Take the two concepts and the wild card your team chose in Unit III
- For each, write the single riskiest assumption in one sentence
- Turn that assumption into a question a prototype could answer this week
- Choose the lowest fidelity that could answer it

ACTIVITY Fifteen minutes. Each team pins up three questions and the fidelity it has chosen for each.

THINK ABOUT IT

What is the riskiest assumption in your team's concept – the one that, if wrong, sinks it?

- Is it about the user, the technology, or the money?
- What would you have to see to believe it was wrong?
- What is the cheapest thing you could build to find out?
- Could you find out this week without building anything at all?

Summary

- A prototype is part of thinking, not the end of it
- Changes made early cost a fraction of changes made late
- Every prototype answers one question you cannot settle by arguing
- Role, look and feel, implementation – name which one you are testing
- Fidelity costs time; go only as high as the question needs
- A prototype you are unwilling to throw away has become a commitment

Next: Sketching and mock-ups

- Why engineers should sketch, badly
- Five marks that draw almost anything
- Thumbnails, sketches and mock-ups
- Storyboarding an experience over time